

(OR)

16. a) Sum the infinite series $\frac{1}{6} + \frac{5}{6.9} + \frac{5.7}{6.9.12} + \frac{5.7.9}{6.9.12.15} + \dots$ (8)

- b) Use Routh criterion to determine the stability of the system represented by the characteristic equation $x^4 + x^3 + 6x^2 + 4x + 2 = 0$. (7)

17. a) Express $\int_0^1 x^m (1-x^n)^p dx$ in terms of Gamma function and hence evaluate $\int_0^1 x^5 (1-x^3)^{10} dx$. (8)

- b) Change the order of integration in $\int_0^1 \int_x^{2-x} \frac{x}{y} dy dx$ and hence evaluate it. (7)

(OR)

18. a) Evaluate $\int_0^{\frac{\pi}{2}} \sqrt{\sin \theta} d\theta$. $\int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{\sin \theta}}$. (8)

- b) Evaluate $\int \int_V dx dy dz$ where V is the finite region of space formed by the planes $x = 0$, $y = 0$, $z = 0$ and $2x + 3y + 4z = 12$. (7)

PART - C

Answer the following

1 x 20 = 20 Marks

19. a) Diminish the roots of the equation $x^4 - 18x^3 + 110x^2 - 266x + 221 = 0$ by 2 and hence solve the equation. (10)
- b) Find the area of an ellipse by using Gamma function. (10)

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

Nov 2025

Course Code: MAT101R01

Course: ENGINEERING MATHEMATICS - I

QP No. :S1663-1

Duration: 3 hours

Max. Marks:100

PART - A

Answer all the questions

10 x 2 = 20 Marks

1. If $\frac{\tan \theta}{\theta} = \frac{2524}{2523}$, then find θ approximately.
2. Prove that $\frac{\sin hx}{1 + \cos hx} = \tanh \left(\frac{x}{2} \right)$.
3. Show that the vectors (1, 3, 2), (5, -2, 1) and (-7, 13, 4) form a linearly dependent set.
4. Two eigenvalues of $A = \begin{pmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -5 & -2 \end{pmatrix}$ are equal and they are double the third. Find them.
5. Determine the nature of the quadratic form $6x^2 + 3y^2 + 14z^2 + 4yz + 18xz + 4xy$.
6. Find the condition that, for the equation $x^3 + px^2 + qx + r = 0$ sum of two roots is equal to zero.
7. Find the value of $\log_e 3 - \frac{1}{2!} (\log_e 3)^2 + \frac{1}{3!} (\log_e 3)^3 - \dots$

8. Evaluate $\int_0^1 x^m \left(\log \frac{1}{x}\right)^n dx$.

9. Change the order of integration in $I = \int_0^2 \int_0^x f(x,y) dy dx$.

10. Find $\int_0^1 \int_0^2 \int_0^3 e^{x+y+z} dz dy dx$.

PART - B

Answer all the questions

4 x 15 = 60 Marks

11. a) If $x = 2\cos\theta$, prove that $\frac{1+\cos 7\theta}{1+\cos\theta} = (x^3 - x^2 - 2x + 1)^2$.

(8)

b) If $\sin(\theta + i\phi) = \cos\alpha + i\sin\alpha$, prove that $\cos^2\theta = \pm\sin\alpha$.

(7)

(OR)

12. a) Express $\cos^4\theta \sin^3\theta$ in a series of sines of multiples of θ . (8)

b) If $i^{i^{\dots\infty}} = A + iB$, prove that $\tan\left(\frac{\pi A}{2}\right) = \frac{B}{A}$ and $A^2 + B^2 = e^{-(\pi B)}$.

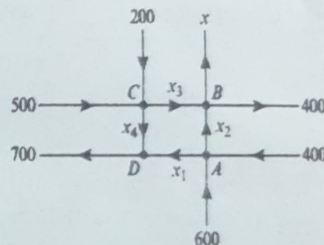
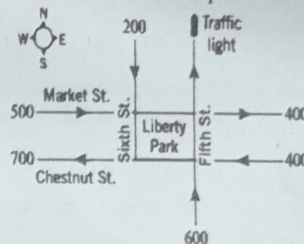
(7)

13. a) The network in figure shows a proposed plan for the traffic flow around a new park that will house the Liberty Bell in Philadelphia, Pennsylvania. The plan calls for a computerized traffic light at the north end of Fifth street, and the diagram indicates the average number of vehicles per hour that are expected to flow in and out of the streets that border the complex. All streets are one-way.

i) How many vehicles per hour should the traffic light let through to ensure that the average number of vehicles per hour flowing into the complex is the same as the average number of vehicles flowing out?

ii) Assuming that the traffic light has been set to balance the total flow in and out of the complex, what can you say about

the average number of vehicles per hour that will flow along the streets that border the complex? (8)



b) Diagonalise the matrix $A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$ by similarity transformation. (7)

(OR)

14. Reduce the quadratic form $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$ to canonical form by an orthogonal reduction and hence find the rank, index, signature of the quadratic form.

15. a) Solve $x^4 - 8x^3 + 7x^2 + 36x - 36 = 0$ given that the product of two roots is equal to the negative of the product of the other two roots. (8)

b) Show that $\frac{1}{1.2.3} + \frac{5}{3.4.5} + \frac{9}{5.6.7} + \dots = \frac{5}{3} - 3 \log 2$. (7)